Internship Learning Reflection Report

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Program:	B.Tech, Artificial Intelligence & ML	**Organization:**	Bairuha Tech
Mentors:	Ambika Raj, Ressaan	**Project:**	FOODFLOW 4.1 – SaaS Food Ordering

1. Introduction and Objectives

During my internship at **Bairuha Tech**, I designed, developed, and deployed FOODFLOW 4.1, a cloud-based multi-vendor food ordering platform connecting customers, restaurant vendors, and administrators. The project bridged academic B.Tech AI & ML knowledge with real-world software engineering practices, providing exposure to enterprise-scale application development, API design, and multi-tenant security structures.

2. Technical Skills Acquired

I gained hands-on full-stack development experience utilizing: **Next.js 16/15** (React 19, TypeScript), **Tailwind CSS v4**, and **Framer Motion** on the frontend; **NestJS 11**, **Node.js**, and **Socket.IO** for backend WebSockets APIs; and **PostgreSQL on Neon Cloud** with **Prisma ORM** for databases. I integrated **Razorpay** (payments), **Resend API** (emails), **Google OAuth 2.0**, and managed deployments using **Vercel** and **Railway**.

3. UI/UX Design and Architecture

We followed a wireframe-first approach to validate user flows before writing code. I designed separate, isolated portal experiences for: **Customers** (menu discovery, live checkout), **Restaurant Admins** (live order board, CRUD, coupons), and **Super Administrators** (merchant approvals, lock accounts, audit logs) to optimize system operations.

4. Key Challenges and Problem Solving

Core challenges resolved during this internship included: **1) Cash on Delivery flow integration:** created a customized checkout bypass skipping the Razorpay transaction overlay while preserving table schemas. **2) Tenant Data Isolation:** structured NestJS Guards matching token claims with route params to prevent cross-vendor metrics crossover. **3) Deployment configurations:** handled broken environment execution pathing directly on the host console.

5. Overall Reflection

This internship provided end-to-end exposure to the software lifecycle—from database normalization and schema design to API implementation, testing, and multi-cloud deployment pipelines. Building FOODFLOW 4.1 has significantly strengthened my software engineering capability, preparing me for complex challenges in full-stack cloud system development.